



Series €FGHE/C



Set-5

प्रश्न-पत्र कोड
Q.P. Code

65(B)

रोल नं.

Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।

Candidates must write the Q.P. Code on the title page of the answer-book.

गणित

(केवल दृष्टिबाधित परीक्षार्थियों के लिए)

MATHEMATICS

(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

निर्धारित समय : 3 घण्टे

अधिकतम अंक : 80

Time allowed : 3 hours

Maximum Marks : 80

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 23 हैं।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में 38 प्रश्न हैं।
- कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें।
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
- Please check that this question paper contains 23 printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains 38 questions.
- Please write down the serial number of the question in the answer-book before attempting it.
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions :

Read the following instructions very carefully and strictly follow them :

- (i) *This question paper contains **38** questions. **All** questions are **compulsory**.*
- (ii) *This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*
- (iii) *In **Section A**, Questions no. **1** to **18** are multiple choice questions (MCQs) and questions number **19** and **20** are Assertion-Reason based questions of **1** mark each.*
- (iv) *In **Section B**, Questions no. **21** to **25** are very short answer (VSA) type questions, carrying **2** marks each.*
- (v) *In **Section C**, Questions no. **26** to **31** are short answer (SA) type questions, carrying **3** marks each.*
- (vi) *In **Section D**, Questions no. **32** to **35** are long answer (LA) type questions carrying **5** marks each.*
- (vii) *In **Section E**, Questions no. **36** to **38** are case study based questions carrying **4** marks each.*
- (viii) *There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 3 questions in Section C, 2 questions in Section D and 2 questions in Section E.*
- (ix) *Use of calculators is **not** allowed.*

SECTION A

This section comprises multiple choice questions (MCQs) of 1 mark each.

1. The domain of the function $\sin^{-1}(2x)$ is :

- (a) $[-1, 1]$
- (b) $[0, 1]$
- (c) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
- (d) $\left(-\frac{1}{2}, \frac{1}{2}\right)$



2. If the matrix $\begin{bmatrix} 0 & 3 & 5 \\ k+1 & 0 & 4 \\ -5 & k & 0 \end{bmatrix}$ is a skew symmetric matrix, then the value of k is :

(a) 4 (b) 2
(c) -2 (d) -4

3. If $A = \begin{bmatrix} 4 & -3 \\ 9 & -3 \end{bmatrix}$, then $|A|$ is equal to :

(a) 15 (b) -42
(c) 0 (d) 25

4. If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ and $A + A' = I$, then the value of α is :

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

5. If A is a 2×3 matrix and B is another matrix such that both $A'B$ and BA' are defined, then order of B is :

(a) 3×2 (b) 2×3
(c) 3×3 (d) 2×2

6. If $y = 3 \log \sqrt{\sin x}$, then $\frac{dy}{dx}$ at $x = \frac{\pi}{4}$ is :

(a) 3 (b) $\frac{2}{3}$
(c) $\frac{3}{2}$ (d) $-\frac{3}{2}$



7. An angle θ , $0 < \theta < \frac{\pi}{2}$, which increases twice as fast as its sine, is :

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

8. The value of k for which the function f , given by

$$f(x) = \begin{cases} kx + 1, & \text{if } x \leq \pi \\ \cos x, & \text{if } x > \pi \end{cases}$$

is continuous at $x = \pi$, is :

- (a) $\frac{2}{\pi}$ (b) $-\frac{\pi}{2}$
(c) $-\frac{2}{\pi}$ (d) $\frac{\pi}{2}$

9. If $y = \sin^{-1}(2x\sqrt{1-x^2})$, $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$, then $\frac{dy}{dx}$ is :

- (a) 2 (b) $\frac{2}{\sqrt{1-x^2}}$
(c) $\frac{-2}{\sqrt{1-x^2}}$ (d) $\sqrt{1-x^2}$

10. The rate of change of the volume of a spherical bubble with respect to its radius r at $r = 3$ cm is :

- (a) $24 \pi \text{ cm}^3/\text{cm}$ (b) $36 \pi \text{ cm}^2/\text{cm}$
(c) $36 \pi \text{ cm}^3/\text{cm}$ (d) $24 \pi \text{ cm}^2/\text{cm}$



11. $\int \cot^2 x \, dx$ equals :

- (a) $-\cot x + x + C$ (b) $-\cot x - x + C$
(c) $\cot x - x + C$ (d) $\cot x + x + C$

12. The value of $\int_{-1}^1 \frac{dx}{5 + 2x + x^2}$ is :

- (a) $\frac{\pi}{8}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

13. The sum of the order and the degree of the differential equation

$$3x^2 \left(\frac{d^2 y}{dx^2} \right)^3 + 5 \frac{d^3 y}{dx^3} + 9 = 0 \text{ is :}$$

- (a) 5 (b) 6
(c) 4 (d) 2

14. A particular solution of the differential equation $dy = y \tan x \, dx$; $y = 1$ when $x = 0$, is :

- (a) $y = 0$ (b) $y = 1 + \sec x$
(c) $y = \sec x$ (d) $y \cos x = 0$

15. If a unit vector $\vec{a}$ makes angles $\frac{\pi}{3}$ with $\hat{i}$, $\frac{\pi}{4}$ with $\hat{j}$ and an acute angle θ with $\hat{k}$, then the value of θ is :

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{6}$



16. The position vectors of two points A and B are $(2\vec{a} + \vec{b})$ and $(\vec{a} - 3\vec{b})$ respectively. The position vector of a point C which divides AB externally in the ratio 1 : 2 is :
- (a) $-3\vec{a} - 5\vec{b}$ (b) $-7\vec{b}$
(c) $\frac{1}{3}(5\vec{a} - \vec{b})$ (d) $(3\vec{a} + 5\vec{b})$
17. The value of λ for which the points A, B and C having position vectors $(3\hat{i} - 2\hat{j} + 4\hat{k})$, $(\hat{i} + \lambda\hat{j} + \hat{k})$ and $(-\hat{i} + 4\hat{j} - 2\hat{k})$ respectively are collinear, is :
- (a) 4 (b) 1
(c) 3 (d) 2
18. The direction ratios of a line whose Cartesian equations are $3x - 3 = 2y + 1 = 5 - 6z$, are :
- (a) 2, 3, -1 (b) 3, -2, 1
(c) 2, 1, -3 (d) 3, 2, -1

Questions number 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c) and (d) as given below.

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).
(c) Assertion (A) is true and Reason (R) is false.
(d) Assertion (A) is false and Reason (R) is true.



- 19.** *Assertion (A) :* The vector equation of a line passing through the points (1, 2, 3) and (3, 0, 2) is

$$\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} - 2\hat{j} - \hat{k}).$$

Reason (R) : Equation of a line passing through two points with position vectors $\vec{a}$ and $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

- 20.** *Assertion (A) :* Given two independent events A and B such that $P(A) = 0.3$ and $P(B) = 0.6$, then $P(A \text{ and not } B) = 0.12$.

Reason (R) : For two independent events A and B, $P(A \text{ and } B) = P(A) \cdot P(B)$.

SECTION B

This section comprises very short answer (VSA) type questions of 2 marks each.

- 21.** (a) Let the relation R be given as

$R = \{(x, y) : x, y \in \mathbb{N} \text{ and } x + 3y = 12\}$. Find the domain and range of R.

OR

- (b) Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^4$ is neither one-one nor onto.

- 22.** Express the matrix $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ as the sum of a symmetric matrix and a skew-symmetric matrix.

- 23.** If $y = x^x$, prove that $\frac{d^2y}{dx^2} - \frac{1}{y} \left(\frac{dy}{dx} \right)^2 - \frac{y}{x} = 0$.



- 24.** Express the vector $\vec{a} = 5\hat{i} - 2\hat{j} + 5\hat{k}$ as the sum of two vectors such that one is parallel to the vector $\vec{b} = (3\hat{i} + \hat{k})$ and the other is perpendicular to $\vec{b}$.

- 25.** (a) Find the angle between the pair of lines given by

$$\frac{x-3}{1} = \frac{y+2}{-2} = \frac{z+4}{2}; \frac{x-5}{-3} = \frac{y+2}{2} = \frac{z}{6}.$$

OR

- (b) If the lines $\frac{x-1}{-3} = \frac{2y-2}{4k} = \frac{3-z}{-2}$ and

$$\frac{x-1}{3k} = \frac{3y-1}{6} = \frac{z-6}{-5}$$
 are perpendicular to each other, find the value of k.

SECTION C

This section comprises short answer (SA) type questions of 3 marks each.

- 26.** Find the intervals in which the function given by

$$f(x) = \sin 3x, \quad x \in \left[0, \frac{\pi}{2}\right] \text{ is (a) increasing (b) decreasing.}$$

- 27.** Find :

$$\int \frac{x-5}{(x-2)^4} e^x dx$$



28. (a) Find :

$$\int \frac{2}{(1-x)(1+x^2)} dx$$

OR

(b) Find :

$$\int \sqrt{3-2x-x^2} dx$$

29. Evaluate :

$$\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$$

30. (a) Solve the differential equation $x \frac{dy}{dx} = y - x \tan \frac{y}{x}$.

OR

(b) Find the particular solution of the differential equation $x \frac{dy}{dx} + y = x \cos x + \sin x$; given $y\left(\frac{\pi}{2}\right) = 1$.

31. (a) Let A and B be the events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{7}{12}$ and $P(\text{not } A \text{ or not } B) = \frac{1}{4}$. Find whether A and B are (i) mutually exclusive (ii) independent.

OR

(b) Find the mean of the number of tails in two tosses of a coin.



SECTION D

This section comprises long answer (LA) type questions of 5 marks each.

- 32.** (a) Let R be a relation in $\mathbb{R}$, the set of all real numbers, defined by $R = \{(a, b) : a \leq b^3\}$.

Show that R is neither reflexive, nor symmetric and nor transitive.

OR

- (b) Let set $A = \{1, 2, 3, \dots, 10\}$ and R be a relation in $A \times A$, defined by $(a, b) R (c, d) \Leftrightarrow a + d = b + c$ for all (a, b) and $(c, d) \in A \times A$. Prove that R is an equivalence relation.

- 33.** Using integration, find the area enclosed by the curve $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

- 34.** While solving the linear programming problem Minimise and Maximise $Z = 3x + 9y$, subject to the constraints $x + 3y \leq 60$, $x + y \geq 10$, $x \leq y$ and $x \geq 0$, $y \geq 0$ graphically, the corner points of the feasible region ABCD are $A(0, 10)$, $B(5, 5)$, $C(15, 15)$ and $D(0, 20)$. Find the minimum value and the maximum value of Z along with the corresponding corner points.

- 35.** (a) Show that the lines $\vec{r} = \hat{i} + 2\hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = \hat{i} + \hat{j} + \hat{k} + \mu(\hat{i} - \hat{j} + 2\hat{k})$ do not intersect.

OR

- (b) Find the coordinates of the foot of perpendicular drawn from the point $(3, -1, 11)$ to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Hence, write the equation of this perpendicular line.



SECTION E

This section comprises 3 case study based questions of 4 marks each.

Case Study – 1

- 36.** A professional typist having his shop in a busy market charges ₹ 200 for typing 8 English and 4 Hindi pages, while he charges ₹ 275 for typing 5 English and 10 Hindi pages.

Based on the above information, answer the following questions :

- (i) If he charges ₹ x for one page of English and ₹ y for one page of Hindi, express the above as a pair of linear equations. 1

- (ii) Express the information in terms of matrix equation $AX = B$. 1

- (iii) (a) Find $|A|$. 2

OR

- (iii) (b) Find $(\text{adj } A)$. 2



Case Study – 2

- 37.** Ravindra started to run a small factory of manufacturing LED bulbs. He can sell x bulbs at a price of ₹ $(300 - x)$ each. The cost price of x bulbs is ₹ $(2x^2 - 60x + 18)$.

Based on the above information, answer the following questions :

- (i) Find the profit function $P(x)$ for selling x bulbs. 1
- (ii) What is $\frac{d}{dx} [P(x)]$? 1
- (iii) (a) How many bulbs should he sell to earn maximum profit ? 2

OR

- (iii) (b) How many bulbs is he selling if he is incurring a loss of ₹ 18 ? 2

Case Study – 3

- 38.** A shopkeeper sells three types of flower seeds A_1 , A_2 and A_3 . They are sold as a mixture where the proportions are 4 : 4 : 2 respectively. The germination rates of the three types of seeds are 45%, 60% and 35%.

Based on the above information, answer the following questions :

- (i) What is the probability of a randomly chosen seed to germinate ? 2
- (ii) What is the probability that the randomly selected seed is of type A_1 , given that it germinates ? 2